



GRADE 1 · INTRO DEBUGGING

Debug Detective

Find the bug, fix it, explain it

Name:

Date:

★ I can find a bug in a longer program, fix it, and explain it.

You are a debug detective

This program is longer, and it has one bug. Run it, find the block that is wrong, fix it, and explain your fix. Then make your own buggy program for a partner to solve. The target is 4.

Bit's number path (target is 4)



The mystery program

start at 0

forward 2

back 1

forward 1

1 Run it

Run the program from 0, one block at a time. What number does Bit land on? Did it reach the target (4)?

2 Find and fix the bug

One block points the **WRONG** way. Circle it, swap it, and run again. What number does Bit land on now?

3 You try — explain and create

Explain why your fix worked. Then write your **OWN** short program that has one bug. Give it a target. A partner can solve it.

Big idea

A debug detective runs the code, finds the bug, fixes it, and checks again. You can find wrong, extra, missing, and out-of-order bugs — you are a debugger!



Debug Detective

Quick facilitation notes & answer key

What this worksheet teaches

The unit finale: students debug a longer program (find the wrong-way block, fix it, explain why the fix works), then make their own buggy program for a partner. Find the bug, fix it, explain it — and make one of your own. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: tape a number line 0 to 5 on the floor and mark the target, so students can walk the buggy program and feel Bit miss.

How to lead it (about 20–25 minutes)

1. Show them (you do). Run the mystery program one block at a time and watch where it goes wrong.
2. Do one together. Exercise 1 — it lands on 2, not 4; Exercise 2 — the wrong-way block is back 1; swap it.
3. Let them try. Students explain why the fix works, then make their own one-bug program for a partner (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — mystery program: 0, forward 2 → 2, back 1 → 1, forward 1 → 2. Bit lands on 2, not the target 4.

Exercise 2 — the wrong-way block is back 1; swap it to forward 1: 0 → 2 → 3 → 4. Bit reaches the target, 4.

Exercise 3 — explanation: the back 1 block moved Bit away from the target; forward 1 keeps Bit heading to 4. Make-your-own is open; a sample is start 0, forward 1, forward 1 with target 3 (a missing-block bug — add one more forward 1).

If students get stuck — and ways to adjust

Watch for: stopping after the fix without explaining why. Have them say what the wrong block did.

Make it easier: walk the mystery program on the floor line.

Make it harder: ask the partner to name the kind of bug before fixing it.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a program, change a block to fix it, and explain what the change did.

You'll know it worked when

The child fixes the mystery program, explains the fix, and writes a solvable one-bug program of their own.